Data Flow Diagram

Purpose

This diagram shows the main information paths in Trip Genius AI: the traveller's trip preferences, backend request validation, AI itinerary generation, persistent trip data, and the separate external map-search flow.

Level 1 Data Flow Diagram

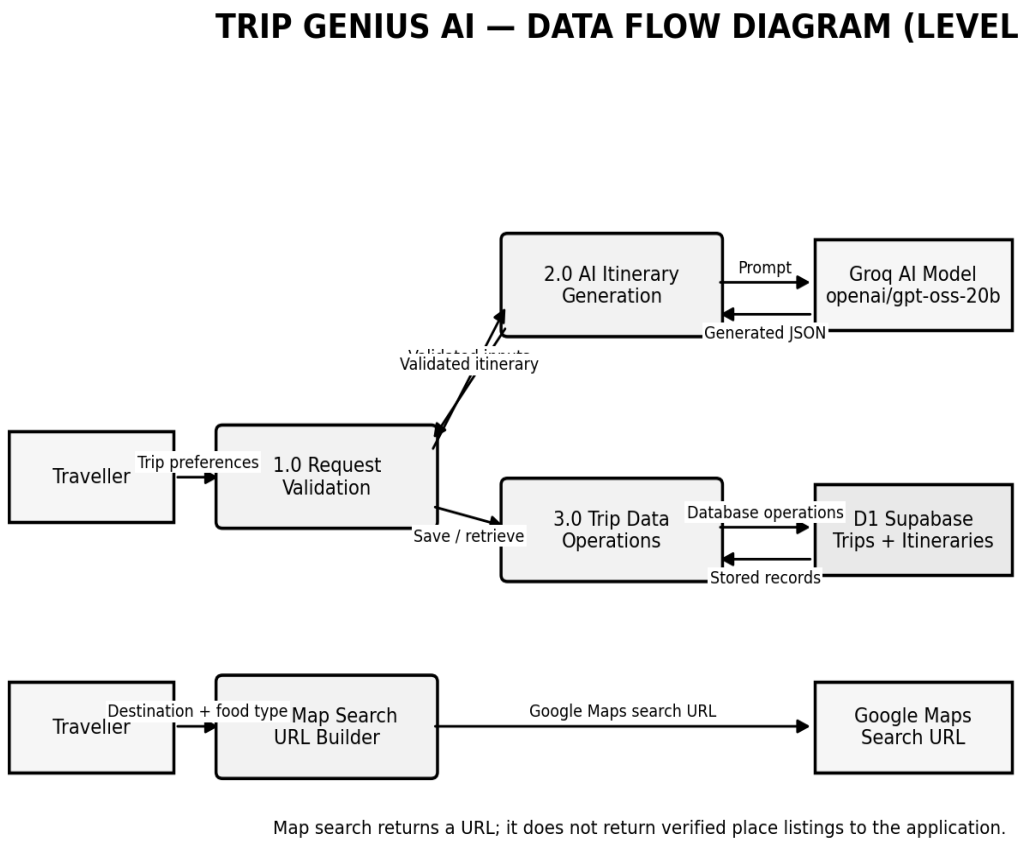


Figure 1. Data flow between the user, API, AI service, database and map-search feature.

Explanation of the flow

1. The traveller enters a starting city, destination, dates, budget, traveller count, interests and travel style in the React frontend. The frontend sends a JSON request to FastAPI.

2. The backend validates the request using the application's data schemas. The AI service builds a prompt and sends it to the configured Groq model. The generated response is parsed and checked for the expected itinerary fields before it is returned.

3. Trip and itinerary information is stored in Supabase. The saved-trip routes list stored trips, retrieve a selected trip, and delete the trip and its associated itinerary. The frontend then displays the response or saved data.

4. Hotel and restaurant endpoints create Google Maps search URLs from the destination and, for restaurants, the selected food preference. The app returns a URL; it does not fetch and verify individual place records.

Data stores, controls and limitations

Supabase is the persistent store for trip and itinerary records. Groq and Supabase credentials must remain in private environment configuration. AI-generated plans and costs are estimates, and map results may vary. The application does not guarantee live prices, availability, opening hours, ratings or dietary suitability.